All-Level Spectral Decimation, Determinants, and the Refinement-Time Boundary on Finite Sierpiński Gaskets

Route-A structural certificate C184

Abstract

For the standard unnormalized Dirichlet graph Laplacian on every finite Sierpiński pre-gasket, we assemble the complete classical 2-, 5-, and 6-series spectrum into one all-level certificate. It includes admissible inverse branches, the forced continuation $6 \mapsto 3$, exact multiplicities and dimension closure, a characteristic-polynomial recurrence, a closed determinant, the heat trace, and the finite spectral zeta. The inverse-branch tree advances graph level rather than physical time; exact finite analytic structure therefore does not pass Route A.

Keywords: Sierpiński gasket; spectral decimation; graph Laplacian; characteristic polynomial; finite spectral zeta.

中文摘要

本文针对任意有限层谢尔宾斯基垫片上的标准非归一化狄利克雷图拉普拉斯算子， 把经典的二、五、六三类谱系整合为一个全层定理：包括可容许反分支、强制步骤 $6 \mapsto 3$、精确重数与维数闭合、特征多项式递推、闭式行列式、热迹和有限谱 泽塔函数。反分支树描述的是图细化层级，而不是固定相空间上的物理时间；因此这些 精确有限维解析结构不能通过路线甲的算术门槛。

关键词：谢尔宾斯基垫片；谱降维；图拉普拉斯；特征多项式；有限谱泽塔。

1 Frozen graph and complete lineage theorem

Let V_m be the level-m pre-gasket vertex set and let $V_0 = \{q_0, q_1, q_2\}$ be the outer boundary. Delete the three boundary rows and columns after forming the full unnormalized graph Laplacian. Thus, on $V_m^\circ = V_m \setminus V_0$,

$$(L_m f)(x) = \sum_{y \sim_m x} (f(x) - f(y)), \qquad f|_{V_0} = 0.$$

Boundary edges still contribute to the diagonal: every interior diagonal entry is four. The full and interior vertex counts are

$$|V_m| = \frac{3^{m+1}+3}{2}, \qquad N_m := |V_m^\circ| = \frac{3^{m+1}-3}{2}.$$

Put

$$R(t) = t(5-t), \qquad \phi_\pm(u) = \frac{5 \pm \sqrt{25-4u}}{2}.$$

Theorem 1 (complete all-level spectrum)**.** *For every $m \geq 1$, the eigenvalues of L_m, with multiplicity, are exactly:*

series	*birth j*	*seed and multiplicity*	*continuation to level m*
2	1	2, *multiplicity* 1	*every inverse word of length* $m-1$
5	$j \geq 1$	5, $a_j = (3^{j-1}+3)/2$	*every word of length* $m-j$
6	$j \geq 2$	6, $b_j = (3^j-3)/2$	*at birth 6; then 3, then both branches*

Here an inverse word means composition of ϕ_- and ϕ_+. If a 6-series is continued beyond birth, the first step is uniquely $6 \mapsto 3$; the algebraic preimage 2 is exceptional and inadmissible in this role.

Local elimination and exceptional kernels. On an old cell with corner values x_0, x_1, x_2, solving the three new midpoint equations for a nonexceptional fine eigenvalue λ gives, cyclically,

$$y_{01} = \frac{(4-\lambda)(x_0+x_1)+2x_2}{(2-\lambda)(5-\lambda)}.$$

Substitution into the old-vertex equations yields $L_m x = R(\lambda)x$; conversely this formula uniquely extends an old eigenfunction when $\lambda \notin \{2,5,6\}$. At the singular values, row reduction of the cell-edge incidence equations gives one level-one 2-mode, new 5-kernel dimension a_j, and new 6-kernel dimension b_j. A new 6-mode restricts to zero. Since

$$R(t) - 6 = -(t-2)(t-3),$$

its next-level extension keeps $t = 3$ and discards the singular $t = 2$ root; both branches resume thereafter. This is the finite-graph decimation and kernel-count argument of Fukushima and Shima [1], to whom the classical theorem and complete spectrum are attributed.

The weighted lineage population is

$$S_m = 2^{m-1} + \sum_{j=1}^{m} 2^{m-j} a_j + b_m + \sum_{j=2}^{m-1} 2^{m-j-1} b_j.$$

The final sum is empty for $m \leq 2$. Since $S_1 = 3 = N_1$ and

$$S_m = 2S_{m-1} - b_{m-1} + a_m + b_m = N_m,$$

induction proves dimension closure and hence completeness.

2 Characteristic polynomial and closed determinant

Write $\chi_m(t) = \det(tI - L_m)$. The base case is $\chi_1(t) = (t-2)(t-5)^2$. For $m \geq 2$, with $b_1 = 0$,

$$\boxed{\chi_m(t) = (-1)^{N_{m-1}}(t-5)^{a_m}(t-6)^{b_m}\frac{\chi_{m-1}(R(t))}{(t-2)^{b_{m-1}}}.} \tag{1}$$

For an old eigenvalue $u \neq 6$, its two preimage factors multiply to $-[R(t)-u]$. For every old 6-mode, division by $t-2$ removes the forbidden preimage and leaves $t-3$. The displayed factorization of $R(t)-6$ proves exact divisibility; the sign makes (1) monic, and the two birth factors add all new modes.

Evaluation at $t = 0$ gives, for $D_m = \det L_m$,

$$D_1 = 2\,5^2, \qquad D_m = 5^{a_m}6^{b_m}\frac{D_{m-1}}{2^{b_{m-1}}}.$$

Separating prime exponents and summing these elementary recurrences proves

$$\boxed{D_m = 2^{(3^m-1)/2}3^{(3^{m+1}-6m-3)/4}5^{(3^m+6m-1)/4}.} \tag{2}$$

The three exponents are nonnegative integers for every $m \geq 1$. Equations (1)–(2) are all-level statements; finite diagonalization is not their proof.

3 Heat trace, finite spectral zeta, and owner boundary

Let $\mathcal{L}_m$ be the lineage set and $q(\ell)$ its birth multiplicity. Because L_m is positive definite,

$$H_m(u) = \mathrm{Tr}(e^{-uL_m}) = \sum_{\ell\in\mathcal{L}_m} q(\ell)e^{-u\lambda_\ell}, \quad \zeta_m(s) = \sum_{\ell\in\mathcal{L}_m} q(\ell)e^{-s\log\lambda_\ell}$$

are exact finite sums. Consequently

$$H_m(0) = \zeta_m(0) = N_m, \qquad -H_m'(0) = \mathrm{Tr}\,L_m = 4N_m, \qquad e^{-\zeta_m'(0)} = \det L_m.$$

The function ζ_m is entire because the sum is finite. This does not select an infinite-gasket normalization or prove a regularized determinant, target functional equation, Gamma factor, divisor, or Weil identity.

A branch word connects different graphs and different Hilbert spaces. It is a refinement genealogy, not an orbit of one autonomous physical-time map on a frozen phase space. Each finite L_m is self-adjoint, so e^{-iuL_m} is a canonical finite unitary group, but its time u is not the refinement level and supplies no arithmetic or target operator. The strict verdict is

$$(A0, A1, A2, A3, A4) = (\text{FAIL}, \text{FAIL}, \text{FAIL}, \text{PARTIAL ANALYTIC STRUCTURE}, \text{FORMAL HINT}),$$

with overall `ROUTE_A_REJECTED`, Route B false, and scope `NO_BAD_EULER_OR_ROOT_NUMBER`.

Exact evidence. Regression evidence contains five levels, 103 lineage rows, 542 exact characteristic-coefficient cells, and 537 graph-eigenvalue cells. An implementation-independent checker passes 3,041 assertions; a separate SymPy path passes 33,177 checks; replay is byte exact; and 71 hostile mutations are rejected. Direct graph work stops at level five and is only a regression sentinel for the written all-level proof.

Revision-round focus. Round 0 freezes the complete 2/5/6 lineage theorem, separates a terminal 6 birth from a continued 6-series, and records the forced $6 \mapsto 3$ boundary.

Edge cases and nonclaims. Level zero has no interior Dirichlet degree of freedom and lies outside the frozen family. Removing boundary rows does not lower interior diagonal degree. A terminal 6 birth remains 6; only a continued series first takes 3. We claim neither priority for spectral decimation, an infinite-gasket spectral zeta, rational-prime semantics, target matching, a Hilbert–Polya operator, Route-B authorization, external peer review, nor a literature-wide novelty certificate.

Declarations. Data/code: package-local evidence and deterministic code accompany the paper. **Ethics:** no human, animal, clinical, personal, or sensitive data; approval is not applicable. **CRediT:** Conceptualization, Formal analysis, Software, Validation, Writing—original draft, and Writing—review and editing; AI systems are not authors. **Funding:** none reported. **Conflicts:** none known. **AI use:** an AI coding assistant supported drafting and exact-code development; it was not an external reviewer or independent error process.

References

[1] M. Fukushima and T. Shima, "On a spectral analysis for the Sierpiński gasket," *Potential Analysis* 1(1) (1992), 1–35. DOI: 10.1007/BF00249784.